Solver backends

PhiFlow
XLB
PICT
FEniCS
INS.jl
OpenFOAM
deal.II
...

Standardized interface

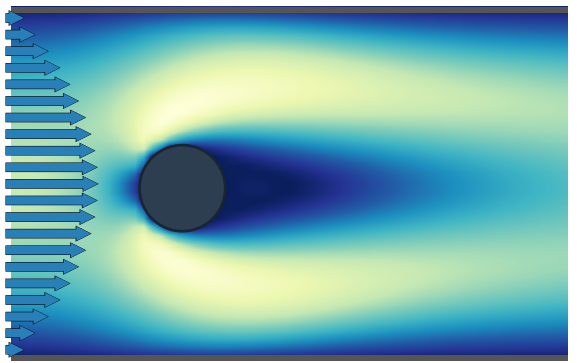
Tesseract

`apply(x) → y`

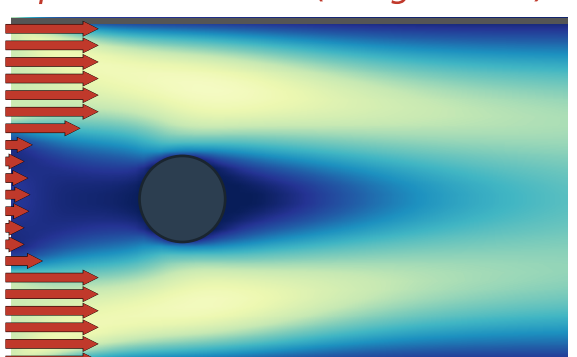
`vjp(x, v) → g`

Benchmark tasks

forward solve



optimized inflow (via gradient)



Evaluation results

2D Navier–Stokes, checks per suite

	Fwd	Cost	Grad	Opt.
JAX-CFD	◑	●	●	◑
PhiFlow	●	●	●	◑
INS.jl	●	●	●	●
XLB	●	●	●	●
PICT	◑	◑	●	●
Warp-NS	◑	●	◑	◑
Exponax	◑	●	●	●
OpenFOAM	◑	●	—	—

● pass ◑ partly — not applicable